Arthur Silva Magalhães

São Paulo, Brazil | arthurmaga@usp.br | arthursmg.github.io | ArthurSMg

Personal Information

- Nationality: Brazilian
- Current position: Ph.D. Student at Universidade de São Paulo

Education

M.Sc. **Universidade de São Paulo**, Astronomy — Mar. 2024 – Mar. 2026
- GPA: 4.0/4.0
- Thesis: Characterization of the young stellar clusters Collinder 205 and Ruprecht 79
- Supervisor: Prof. Jane Gregorio-Hetem

B.Sc. **Universidade de São Paulo**, Physics — Apr. 2021 – Dec. 2023
- GPA: 8.1/ 10.0
- Undergraduate Research: Near-Infrared study of regions associated with young star clusters
- Supervisor: Prof. Jane Gregorio-Hetem

B.Sc. **Universidade Federal de Minas Gerais**, Physics — Feb. 2019 – Feb. 2021 (*transferred*)
- Undergraduate Research: Astrophysical parameters and evolution of star clusters
- Supervisors: Prof. Wagner Corradi, Prof. João Francisco C. Santos Jr.

International Experience

Exchange Semester

Uppsala University, Sweden — Jan. 2025 - Jun. 2025

- Research Projects: Testing webSME for abundance determination; Probing NGC 6397 for Atomic Diffusion using webSME
- Supervisor: Prof. Andreas Korn
- Courses: Nuclear Astrophysics; Astrophysical Tests of Physical Theories (*Passed with distinction*)

Scholarships

CAPES — 2024 - 2026
Coordenação de Aperfeiçoamento de Pessoal de Nível Superior
Master of Sciences Scholarship

Erasmus+ — 2025 - 2025
International Credit Mobility Program
Full-time exchange studies

CNPq — 2022 - 2023
Conselho Nacional de Desenvolvimento Científico e Tecnológico
Undergraduate Research Scholarship

CNPq — 2020 - 2021
Conselho Nacional de Desenvolvimento Científico e Tecnológico
Undergraduate Research Scholarship

Publications

Conference Proceedings

Characterization of the young stellar cluster Collinder 205 with Spartan/SOAR 2025
A.S. Magalhães, J. Gregorio-Hetem
Boletim da Sociedade Astronômica Brasileira

Near-infrared study of regions associated with young star clusters 2024
A.S. Magalhães, J. Gregorio-Hetem
Boletim da Sociedade Astronômica Brasileira

Professional & Observing Activities

IAG - USP, *33°International Symposium on Scientific and Technological Initiation at USP* 2025
Poster evaluator

IAG - USP, *32°International Symposium on Scientific and Technological Initiation at USP* 2024
Poster evaluator

Observatório Pico dos Dias (OPD), *Multiple Observing runs* 2019 - 2026
1.6 m Perkin-Elmer Telescope and 0.6 m Boller & Chivens Telescope (photometry and spectroscopy).

Southern Astrophysical Research Telescope (SOAR), *2 nights* 2026
Goodman High Throughput Spectrograph and SOAR Adaptive Optics Module (SAM).

Participation in Events

Presenter

Talks

X IAG Science Day 2025
São Paulo - Brazil
Using webSME to study atomic diffusion in the globular cluster NGC6397

IX IAG Science Day 2024
São Paulo - Brazil
Characterization of the young stellar cluster Collinder 205 with Spartan/SOAR

VIII IAG Science Day 2023
São Paulo - Brazil
Astrophysical Parameters Determination for Young Star Clusters

Posters

11th meeting of the BRICS Astronomy Working Group 2025
São José dos Campos - Brazil
Probing atomic diffusion in the globular cluster NGC6397 using VLT/FLAMES-UVES data

XIX IAG/USP Advanced School on Astrophysics 2025
Bertioga - Brazil
Testing webSME for abundance determination & probing atomic diffusion in NGC 6397.

XLVII Annual Meeting of the Brazilian Astronomical Society 2024
Águas de Lindóia - Brazil
Characterization of the young stellar cluster Collinder 205 with Spartan/SOAR

31°International Symposium on Scientific and Technological Initiation at USP 2023
São Paulo - Brazil
Near-infrared study of regions associated with young star clusters

XLVI Annual Meeting of the Brazilian Astronomical Society 2023
Rio de Janeiro - Brazil

Near-infrared study of regions associated with young star clusters

XXIX Undergraduate Research Week at UFMG 2020
Belo Horizonte - Brazil
Study of the interstellar cloud TGUH1821 while searching for open clusters

Teaching Assistant

Instituto de Astronomia, Geofísica e Ciências Atmosféricas da USP 2024
AGA0414 - Methods in Observational Astrophysics I

Technologies

- **Programming Languages:** Python, R
- **Astronomical Data:** *Gaia*-DR3, AllWISE/NEOWISE, 2MASS
- **Tools:** Linux, LaTeX, git, webSME, Topcat, Aladin, DS9, StarFinder, `IRAF`

Outreach

Science Communication

Astronomia ao meio-dia 2026 - present
Weekly department outreach seminars (organizer)

Astrobites 2026
Guest post

Astropontos 2025 - present
Regular author (The PT-BR equivalent of Astrobites)

Creator, independent Youtube Channel 2020 - present
Videos about life in academia, astrophysics outreach and software tutorials

Invited Talks

Star Clusters in the Gaia Era: Did You Know Stars Have Families Too? 2026
Astronomia ao meio-dia - IAG/USP, São Paulo

Star Clusters: the Jewels of the Galaxy 2026
IntegraAstro 2026 - IAG/USP, São Paulo

Telescope Proposals

- SOAR Telescope (Goodman - Long slit, 2026B) — **Principal investigator** on a time proposal to obtain a follow up optical spectra of variable sources in the Collinder 205 open cluster and the CMa star-forming region. (*In analysis*)
- SOAR Telescope (SAM - Adaptive Optics Module, 2026B) — **Co-author** on a time proposal to gather follow-up observations to characterize the interstellar medium of the targets in the previous proposal. (*In analysis*)
- SOAR Telescope (SAM - Adaptive Optics Module, 2026A) — **Co-author** on a time proposal to observe two star clusters and four star-forming regions. (*Accepted*)
- SOAR Telescope (Goodman - Multi Object Spectroscopy, 2026A) — **Co-author** on a time proposal to obtain optical spectra of variable sources in the Collinder 205 open cluster and the CMa star-forming region. (*Accepted*)
- T-80 South Telescope (S-PLUS filters, 2026A) — **Co-author** on a time proposal for variability analysis of four star-forming regions. (*Accepted*)
- OPD 1.6 m Perkin-Elmer Telescope (Coudé spectrograph, 2026A) — **Co-author** on a time proposal to obtain optical spectra of variable sources in the Collinder 205 open cluster and other star-forming regions. (*Accepted*)

Other Information

- **Languages:** Portuguese (Native), English (Proficient), Spanish (Intermediate), German (Basic)
- **Work in progress: Magalhães, A. S.** & Gregorio-Hetem, J. (in prep. for submission to Astronomy & Astrophysics). *Revisiting the Young Stellar Populations of Collinder 205 and Ruprecht 79 with Gaia DR3 and Near-Infrared Data*

References

- **Prof. Jane Gregorio-Hetem**
  Department of Astronomy, Institute of Astronomy, Geophysics and Atmospheric Sciences, University of São Paulo
  Email: gregorio-hetem@usp.br
  Phone: +55 11 3091-2823

- **Prof. Phillip Galli**
  Department of Astronomy, Institute of Astronomy, Geophysics and Atmospheric Sciences, University of São Paulo
  Email: phillip.galli@iag.usp.br
  Phone: +55 11 3091-2718